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Passion and Strategy: Novozymes' Embrace of the UN Sustainable Development Goals

The United Nations Sustainable Development Goals (UN SDGs) express in a good way some of the major challenges that the planet and mankind will be facing in the next decades. We certainly have pride in making these goals our obligations. But solving some of these issues has to carry significant rewards. It's not only about doing good. It's about doing good and being rewarded for it.

— Peder Holk Nielsen, president and chief executive officer (CEO), Novozymes A/S

On a sunny morning in June 2017, just north of Copenhagen, Denmark, Peder Holk Nielsen, president and CEO of Novozymes A/S, reversed into his parking space, switched off the engine, unbuckled his seat belt, and stepped out of his car. As he walked from the parking lot towards his office, Nielsen could not stop thinking about his company and the challenges that lay ahead. In a few days, he would have to make a recommendation to the Novozymes board of directors on the firm's sustainability strategy. He was faced with a difficult choice.

With 6,500 employees, sales of DKK 14.14 billion (about \$2.01 billion),^a and a commanding 48% market share as of 2016,¹ Novozymes was the world's largest and oldest producer of industrial enzymes. Serving more than 40 industries, including biofuels, detergents, food, and animal feed, Novozymes helped its customers "make more from less" by offering biological solutions that produced less waste, reduced the use of chemicals, and saved energy and raw materials. By definition, enzymes made reactions more efficient and faster. As such, Novozymes had a compelling reason to integrate sustainable principles into its operations.

Novozymes had consistently been recognized for its leadership in sustainability, notably among policymakers, civil society, and peers. In 1993, Novozymes^b had been the first company in Denmark to issue an environmental report, and one of the first to implement the "triple bottom line" aiming to balance financial, social, and environmental aspects.² In 2015, Novozymes became the world's first company known to have crafted a new corporate strategy based on the United Nations Sustainable

^a Financials translated into USD at an exchange rate of 0.14 USD/DKK.

^b As a former division of Danish company Novo Nordisk.

Senior Lecturer Andy Zelleke and Research Associate Emilie Billaud (Europe Research Center) prepared this case. It was reviewed and approved before publication by a company designate. Funding for the development of this case was provided by Harvard Business School and not by the company. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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Development Goals (SDGs). Since then, the company had used the SDGs as a lens to screen products, business models and partnerships, and had prioritized innovation decisions that could deliver extraordinary societal and business impact.³

Yet some of these bets were risky and the R&D outcomes were unpredictable. Novozymes was also facing uncertainty—regulatory, and in public attitudes—as the company grappled with problems of enormous and enduring consequence, such as forging climate change policy or managing stakeholder expectations related to environmental sustainability. Finally, 2016 had been a year with more headwind than usual. Low raw material prices had dampened customer demand for cost-saving enzymatic solutions.⁴ As a result, the firm's revenues had grown at a much slower pace than expected, resulting in layoffs to safeguard profitability.

Given the financial pressures on the company, Nielsen knew that any decisions he took to the board would need a strong growth rationale. He reflected on the past few years, and debated what the future held for sustainability at Novozymes. Consequential decisions had to be made to reprioritize the company's efforts. Should the firm stay the course on its present sustainability journey? Should it double down on its commitment to the SDGs? Or was it time, perhaps, for a change of course—and if so, in which direction?

Company Background

Company History

The history of Novozymes spanned more than 95 years. Its roots dated to the 1920s, when *Nordisk Insulinlaboratorium* and *Novo Terapeutisk Laboratorium* were established to develop, manufacture, and sell insulin in Copenhagen, Denmark.⁵ Both firms rapidly expanded and competed to be the first on the market with world-class diabetes medicine.⁶ While the founders of *Nordisk Insulinlaboratorium* belonged to the Danish intellectual elite with deep roots in the world of science, the founders of *Novo Terapeutisk Laboratorium* compensated for what they lacked with notable entrepreneurial spirit.⁷

Over the years, the rival firms began to diversify by developing other products. *Nordisk Insulinlaboratorium* developed drugs for the treatment of hemophilia and growth disorders, while *Novo Terapeutisk Laboratorium* became the world's largest producer of industrial enzymes.⁸ In 1989, despite years of intense competition, both firms decided to join forces under the name Novo Nordisk A/S.⁹ The combined company was estimated to control up to half of the market for insulin in the Western world.¹⁰ The merger also brought the establishment of the Novo Nordisk Foundation—an independent institution whose objectives were to create a solid basis for the operation of Novo Nordisk and to support scientific, humanitarian, and social causes.¹¹

In the early 1990's, when the term "triple bottom line"^c was first coined, Novo Nordisk committed itself to implement this philosophy.¹² In 1993, it was the first company in Denmark, and one of the first in the world, to prepare an environmental report describing its resource consumption, emissions, and use of experimental animals. In 1999, Novo Nordisk also published its first social report, documenting the company's efforts to live up to its social responsibility towards employees, patients, communities, and other stakeholders.¹³

^c The Triple Bottom Line was a method of evaluating corporate performance by measuring profits, as well as environmental sustainability and social responsibility.

In 1999, Novo Nordisk underwent a restructuring. Having developed into two very different businesses—pharmaceuticals and industrial enzymes—the firm decided to demerge the enzyme business from Novo Nordisk, giving the new entity the name Novozymes A/S.¹⁴ The demerger enabled the two resulting firms to increase their operational freedom and focus on what each did best. In 2000, Novo Nordisk and Novozymes began operating as two separately listed companies.¹⁵

Early Sustainability Journey

Since the demerger in 2000, Novozymes had passed through several stages of growing commitment to sustainability. (See **Exhibit 1** for more details on the three major stages of Novozymes' sustainability journey.)

2000–2008: Manage Reputation. Initially, sustainability was primarily a matter of compliance, cost reduction, and risk avoidance. It was perhaps also, in part, a public relations effort at keeping potential critics at bay.¹⁶ Sustainability activities were focused on controlling operational and supply chain risks, reducing costs, and increasing transparency and reporting. For example, in 2002, Novozymes became the world's first known firm to produce an integrated financial, environmental, and social report.¹⁷ Publishing a unified report of this type was the natural consequence of business and sustainability moving ever closer together, and of various stakeholders asking for a wider overview of the business.¹⁸ In 2003, Novozymes also launched a program called "Purchasing with Decency." This program was primarily devised to draw its suppliers' attention to the importance of ensuring compliance with basic human rights and labor standards. Finally, in 2005, the firm announced six integrity principles to help individual employees deal with challenges such as bribery pressures. While these first sustainability initiatives were modest, they were still significant, as the process of engaging employees and heightening their sustainability consciousness had begun.¹⁹

Claus Stig Pedersen, head of corporate sustainability affairs, joined Novozymes in 2006. With a M.Sc. in Chemical Engineering and a Ph.D. in Sustainability Management, Pedersen had been actively involved in environmental protection for most of his life. In his early twenties, he had joined Greenpeace and left-wing political parties, believing that the ecological crisis he perceived would require radical changes in attitudes and lifestyles. He soon came to believe that business could also play a useful role in making the world a better place. Pedersen recalled:

During my studies, I realized that businesses had to be part of the solution. I quickly got interested in life cycle assessments (LCAs).^d I thought that by looking at the whole value chain instead of just an individual company, I would understand the system and there would be opportunities to find positive drivers of change. I got very excited by this idea. I met pioneers in LCAs, who had received grants from the Danish government to develop methodologies. Their work was groundbreaking. I joined their team and as an LCA consultant, I developed practical and value-adding ways to integrate sustainability into a variety of business functions from sourcing and R&D, to sales, marketing, and investor relations. After graduation, I was hired by one of my clients—a packaging company—and I helped transform this unknown player into a rock star in sustainability.

Recognized as an early leader in LCA methodologies, Pedersen was approached by Novozymes in 2006. Pedersen recalled, "When I joined the firm, I followed strong environmental, health, and safety (EHS) guidelines; I built and maintained a high level of stakeholder engagement; and Novozymes was

^d LCAs were a set of procedures for compiling and examining the inputs and outputs of materials and energy and the associated environmental impacts directly attributable to the functioning of a product or service throughout its life cycle.

acknowledged by indices and awards for its transparency. However, Novozymes publicly announced that its ambitions were not about sustainability. Its stated ambitions were to achieve revenue of DKK 10 billion by 2010." He added, "The company was certainly acting very responsibly and had high standards when it came to interacting with people, protecting the environment, reducing waste, making reports, and building the reputation of a responsible firm. However, the organization failed to capture potential value from environmental, social, and governance issues." Novozymes' slogan—"unlocking the magic of nature"—was somewhat passive.²⁰ Looking back at this period, Novozymes' current CEO, Peder Holk Nielsen, expressed a similar view to Pedersen's, "We didn't really see sustainability as a driver of growth."

2009–2014: Support Current Business. In 2008, prices of raw materials rose dramatically. Novozymes' earnings, like those of most other companies, were also affected. The firm relied on its technology to minimize the adverse effects through productivity optimization or raw material substitution.²¹ Over time, the firm gradually started talking about how sustainability could become a vehicle for revenue and profit growth. Claus Crone Fuglsang, senior vice president, research and technology, explained, "Initially it wasn't a big change. We were just growing up as a company, realizing what our products could do in terms of environmental footprint."

As the years went by, the company progressively started setting more ambitious targets to further integrate sustainability into its business strategy. Pedersen said, "With our new vision 'Rethink Tomorrow', our ambition was nothing less than 'changing the world'. We started working really hard to integrate sustainability in every corner of the business: from sales, to production, marketing, stakeholder relationship, or communication. We generated significant resources during that period."

Novozymes' sustainability focus shifted from supply chain and operations to products and customers. Product LCAs were conducted in collaboration with customers to document the sustainability benefits generated by Novozymes' solutions—which customers could use in their own marketing activities. LCAs also formed the basis for communication and corporate storytelling about the potential of Novozymes' technologies and for dialogues with investors, policymakers, and NGOs.

The company also created an executive sustainability board, which included vice presidents of each business function. Giving each member direct responsibility for results helped ensure that efforts to meet sustainability targets also created value for the business.²² In addition, a portion of annual employee bonuses was tied to meeting the firm's sustainability goals.²³ As the company began to become more knowledgeable about sustainability principles and approaches, the scope of the economic benefits started to become apparent.

A New Strategic Plan Inspired by the UN Sustainable Development Goals

The ultimate evolution for Novozymes was for sustainability to be seen as a way to create new business segments, products, and growth. As more functional units were engaged in the sustainability process, it was becoming apparent that the company's functional, siloed efforts to date could be better aligned strategically for greater impact and efficiency.

When Peder Holk Nielsen replaced the retiring Steen Riisgaard as president and CEO of Novozymes in 2013, he brought deep company experience to this strategic opportunity. Nielsen's background was both in chemical engineering and business management, and his 28-year career at Novozymes spanned positions across business development, research and development, quality management, and sales and marketing. Nielsen said, "My longevity in the company, going all the way up to CEO, is not typical, but it's not incredibly unusual either. I took on a lot of different tasks and

jobs at the firm. I improved the efficiency of many processes and with the help of our team, I fully embraced sustainability.”

Within a year of Nielsen's appointment as CEO, Novozymes' leadership was looking into the future and identifying trends and issues against which to align the firm's organizational priorities. The executive leadership team and Pedersen, the head of sustainability, were asked to provide input to the strategic planning process, so that a new vision statement and a new strategic map could be defined. Pedersen said, “Of course, sustainability was part of the discussion. We were doing really well, so we started thinking about what else could be done. What comes after ‘changing the world’?”

The information collected during this assessment phase helped Nielsen and his team formulate a new long-term strategy for the company. New goals and priorities were initially presented to the board of directors during a meeting in Italy in 2014. The board pushed for a still more ambitious strategy. Agnete Raaschou-Nielsen, vice-chairman of the board, recalled, “We thought that this new plan was quite underwhelming. It did not respect the fact that Novozymes was such a special company. Our recommendation was for the executive team to come back with a new plan that would reflect the essence of Novozymes in a much better way.”

Meanwhile, societies around the world were under pressure due to environmental degradation and social tensions. In 2000, 189 countries had gathered at the United Nations headquarters to sign the Millennium Declaration, in which they had committed to achieving a set of eight measurable goals (the Millennium Development Goals, or “MDGs”) that ranged from halving extreme poverty to promoting gender equality, by the target date of 2015.²⁴ With the MDG target date fast approaching, discussions of a post-2015 agenda were initiated. In 2012, the world had gathered again in Rio de Janeiro, Brazil, for the United Nations Conference on Sustainable Development, or ‘Rio+20’. An outcome of this conference had been a commitment to creating a new set of goals: the United Nations SDGs, which would extend the momentum created by the MDGs. While the MDGs had been determined by a small team of technical experts, the SDGs had been developed by a 30-member Open Working Group (OWG) under the auspices of the United Nations General Assembly (UNGA).²⁵ The group had met regularly, and many of the 30 seats had been shared between several countries. Civil society had also been greatly involved in this process.²⁶ The work of the OWG was completed in mid-2014, and a set of 17 proposed goals and 169 proposed targets were presented in September 2014.²⁷ (See **Exhibit 2** for more details on the 17 proposed goals.)

Novozymes had been engaged in the SDG process since its very beginning, and had actively advocated for an SDG framework with a prominent role for business. Nielsen explained, “It's clear that the private sector plays a key role and bears a big responsibility to drive the global sustainability agenda, and the SDGs provide a signpost that points companies in the right direction.” Accordingly, in the fall of 2014, Novozymes' executive leadership team reformulated its long-term strategy and purpose, using the draft SDGs as inspiration. Board member Raaschou-Nielsen recalled, “The CEO and his team got back to us and explained that Novozymes should stand by the SDGs. The board was thrilled to see that it was possible to develop such an interesting vision.”

In January 2015, Novozymes' “Partnering for Impact” strategy was introduced, setting out four focus areas that would enable the firm to fulfill its purpose: (1.) Rallying for change: forming partnerships and networks with customers, consumers, governments, suppliers, and academia to make a sustainable difference. (2.) Leading innovation: exciting consumers by delivering significant innovation, tailored to local markets. (3.) Focusing on opportunities: prioritizing the customers, markets, and activities that held the biggest opportunities for creating impact. (4.) Growing people: supporting Novozymes' employees and people around them in unfolding their full potential.²⁸

Novozymes also identified six long-term sustainability targets to fulfill the company's purpose. These ambitious targets included reaching six billion people with its biological solutions by 2020, educating one million people about the potential of biology from 2015 to 2020, catalyzing five global partnerships from 2015 to 2020, delivering ten transformative innovations from 2015 to 2020, saving 100 million tons of CO₂ by 2020, and enabling Novozymes' employees to develop by 2020.²⁹

This new strategic map led to the dissolution of Novozymes' sustainability board, which previously had been the governing body for Novozymes' sustainability-related activities. From this point on, sustainability was governed at the very top level of Novozymes—its board of directors and executive leadership team—since its commitment was firmly established as a top business priority.³⁰

In March 2015—six months before the SDGs were officially signed by 193 countries—Pedersen appeared in front of a group of negotiators and NGOs at the United Nations headquarters in New York City. He explained to his audience how Novozymes planned to implement the SDGs:

Our plan was to break down the global goals into sub-topics and define operational metrics: some goals would be quantified; some would be qualified; some wouldn't have any impact on us. We were going to use this process to make decisions and prioritize our innovation pipeline. The audience was shocked. They replied: 'This is what we have been dreaming about. But we are not even done yet with the SDGs. How can you move so fast?' They were excited that our business embraced the SDGs and at the same time, they were stunned. It was 6 months before the SDGs were signed, but there were some solid drafts. This was more than enough for us to act. That gave us a very strong leadership position in terms of embracing and using the SDGs for strategic development and management of the company. We have been standing on that wave ever since.

2015–2020: From Commitment to Action. Sustainability was managed by two teams, corporate sustainability and sustainability business development. Corporate sustainability worked to ensure that Novozymes was able to influence the global sustainability agenda while conducting sustainability trend, stakeholder and materiality analyses, integrating learnings into the business, and reporting on progress. Sustainability business development worked with a range of business units to help identify and leverage sustainability drivers for business growth, including conducting LCAs to expand product value propositions.³¹

Prioritizing Innovation Decisions

In the next period (2015–2020), the company believed that business and sustainability needed to effectively merge. Novozymes' intent was to leverage sustainability insights to identify and prioritize the business opportunities that could deliver societal and business impact, and to leverage its sustainability leadership position to accelerate and grow those opportunities. Nielsen commented, "For some activities, the SDGs represented a confirmation of things that we were already doing. But there were also areas that we had not considered before, because we did not comprehend that this was a significant challenge for mankind as we moved towards 2030." Pedersen added:

My hope is that we can use the SDGs as a crystal ball looking into the future. We need to identify global challenges that represent opportunities for our technology. We will then work backwards from there—creating technology and business model solutions to deliver on these needs. Our leadership position is based on our technology, our strategic thinking, but also our execution. Even with the most brilliant strategy, the battle is only half won. We need to fully deliver on our plans and make the SDGs operational. We need to put these good intentions into action.

Raaschou-Nielsen elaborated, "The board was extremely active in saying: "This is great, but how do we transform the SDGs into goals? How do we make sure that these goals and the KPIs that emerge from this are issues that could be audited in the future?"

To qualify and quantify potential contributions to the SDGs stemming from its business activities, Novozymes developed an assessment and management tool, called the "Global Goals Assessments." The company identified 15 quantifiable and operational impact categories most relevant to its business, such as food supply, sanitation, and waste. (See **Exhibit 3** for details on these categories.) It also updated its operational targets, such as energy efficiency, reduction in CO₂ intensity, employee satisfaction and motivation, and women in senior management. (See **Exhibit 4** for a list of targets.)

The company established a new Portfolio Board to manage the R&D pipeline across Novozymes' divisions. This board used the Global Goals Assessments tool to scan its innovation pipeline and identified the most potent future contributors to the SDGs: the "Globe Star Projects." (See **Exhibit 5** for more details on this process.) Innovation opportunities were evaluated and prioritized based on their market potential, strategic fit, and Global Goals impact potential. Andrew Fordyce, executive vice president food and beverage, said, "We have financial and non-financial overarching targets that we are shooting for. We are looking at how we could achieve success in those areas on aggregate."

Novozymes was in the process of learning how it could use this information to attract new types of partners with capacity to help accelerate its development. Four high impact commercial projects, in animal feed, bioenergy, palm oil, and water, were in scope within that process. Thomas Videbæk, chief operating officer and executive vice president for research, innovation, and supply, added, "For each innovation, we are making a triple bottom line calculation. We look at criteria, such as: how fast the technology could be brought to market. When it's time to make a decision, it's a question of balance. There is always an economic parameter of course, but we are willing to moderate our risk aversion if the sustainability impact is significant. Some projects wouldn't have made it under normal scrutiny, but they went ahead anyway because they had the potential to be transformational."

Preliminary Results

In 2017, Novozymes continued to use LCAs to assess the environmental impacts of its solutions covering the entire life cycle from cradle to grave—from raw material extraction, through production, use, and transport, to final disposal, or recycling. Based on these studies, Novozymes estimated that in 2016 its customers avoided an estimated 69 million tons of CO₂ emissions by applying its solutions.³² The savings achieved were equivalent to taking 30 million cars off the road.³³ Nielsen said, "I am proud of this achievement, but at the same time, we were unable to meet some of our other sustainability targets, for example water efficiency, frequency of occupational accidents, and diversity. This is not satisfactory, and in 2017 it will be key for us to deliver a better performance."³⁴

Novozymes was dedicated to optimizing its operations year on year, so as to reduce the consumption of natural resources and mitigate the negative environmental impact of its production processes. In 2016, its energy efficiency improvement (compared with the base year 2014) was 10%. This was short of the 2016 target of 18%, primarily due to slower-than-expected gross profit development.³⁵ In 2016, Novozymes also obtained 24% of its energy from renewable sources, mostly from offshore wind farms in Denmark.³⁶ The water efficiency improvement (compared with the base year 2014) was 6%. Enzyme manufacturing was a water-intensive process that also generated significant amounts of wastewater. Novozymes was striving to use water more efficiently.³⁷

In 2016, Novozymes' production sites across the world generated 13,000 tons of solid waste, of which 44% was recycled. Novozymes also recovered more than 97% of the biomass generated as a by-

product of its production processes.³⁸ Novozymes' management approach to waste and by-products was site-specific. This was because waste handling was a complex issue that was regulated locally and involved many external service providers. (See **Exhibit 6** for more environmental data.)

From a social perspective, Novozymes continued to strengthen diversity within the organization while upholding the principles of merit-based selection and promotion. In 2016, 36% of the employees promoted were female. Novozymes' employee absence rate remained at 2.0%. Region-specific initiatives had been drawn up to reduce absences.³⁹ The frequency of occupational accidents decreased to 2.2 per million working hours from 2.5 per million working hours in 2015. As most accidents involved falls or trips, a new program had been launched across all sites to raise awareness and reduce these types of accidents.⁴⁰ Finally, in 2016, 99% of Novozymes' employees had completed business integrity training.⁴¹ (See **Exhibit 7** for more social and governance data.)

Novozymes also measured the success of its purpose and strategic focus areas using its six long-term targets. For example, Novozymes encouraged local education with global impact by driving EDUCATE activities out of its largest regions: China, India, Brazil, North America and Europe, Middle East and Africa. In 2016, the company educated more than 100,000 learners through regional partnerships. This brought its total to more than 130,000 learners since 2015 – for a target of one million people by 2020.⁴² (See **Exhibit 8** for an update on the firm's achievements.)

Finally, the firm received the prestigious Best Business Partner of the Year Award from P&G for the seventh time and it was recognized for its consistent high-level performance with their Excellence Award. Its continuous sustainability efforts also won recognition in 2016. Once again, the company scored 90 out of 100 in the chemicals sector of the Dow Jones Sustainability World Index, placing it among the top companies in terms of sustainability in the competitive chemicals sector.⁴³ The United Nations Global Compact also recognized Pedersen as a local SDG Pioneer for his work to align business with the SDGs. Pedersen was the only European business representative to be honored.⁴⁴

Company Overview as of 2017

As of 2017, Novozymes remained headquartered in Bagsværd, northwest of Copenhagen, Denmark, with production sites in Argentina, Brazil, Canada, China, Denmark, England, India, and the United States. The firm produced more than 700 kinds of enzymes, microorganisms, biopolymers, and proteins used by 40 industries in about 140 countries.⁴⁵ Enzymes were proteins that catalyzed biochemical reactions and were derived from microorganisms such as bacteria, fungi, and yeasts. They were produced on an industrial scale by fermenting production organisms (strains) developed in laboratories. With a few exceptions, the firm had introduced every new enzyme to the industry.⁴⁶

Novozymes sold products that entailed savings of various kinds. Enzymes proved effective in improving product performance: for example, food enzymes extended the freshness of bread, and detergent enzymes broke down stains in lower washing temperatures. Enzymes also improved production yields, reducing the per unit input of raw materials whose prices were often volatile. Additionally, enzymes improved the sustainability of production as they shortened manufacturing processes and required less energy and water to operate. Finally, compared with conventional chemicals, enzymes' main substitute, the latter were biodegradable and reduced pollution.⁴⁷ (See **Exhibit 9** for more details on the roles of enzymes and microorganisms.)

COO Videbæk remembered his first project at the firm, "In the 1980s, stonewashed jeans were popular. To give this worn-out appearance, jeans used to be placed in large clothes washers that were also filled with large stones. As the wash cylinder rotated, the cloth fibers were repeatedly pounded

and beaten by the tumbling stones. The noise was incredible." He added, "Novozymes developed a cellulose enzyme that had an effect on jeans. For a year, I set up tests and tried to find correlations. It was a fantastic job. During our first trial in Spain, we replaced stones by a liter of enzymes; the end result was amazing. Employees were crawling into the washers, thinking that this was a trick."

Novozymes' sales in 2016 were DKK 14.14 billion, reflecting only 1.0% growth from 2015. (See **Exhibit 10** for key financials and **Exhibit 11** for historical financial performance.) Although Novozymes A/S was based in Denmark, it derived most of its sales outside of its home market. In 2016, 36% of sales were generated in Europe, Middle East, and Africa; 34% in North America; 20% in Asia Pacific; and 10% in Latin America.⁴⁸ Despite its disappointing revenue growth in 2016, Novozymes remained highly profitable (with a profit of DKK 3.05 billion) and continued to generate impressive free cash flow.⁴⁹ The firm had used debt (and some equity) to make several strategic acquisitions in recent years. Novozymes had also returned value to shareholders through dividends and share buybacks. For 2016, the proposed dividend was DKK 4, a 14% increase from the previous year and a 39% payout ratio. The firm also instituted a share buyback program to purchase up to DKK 2 billion in shares over the course of 2017.⁵⁰ Novozymes' long term financial targets included an EBIT margin above 26% and a return on invested capital (ROIC) over 25%.⁵¹

In February 2016, Novozymes had also changed its organizational structure and had appointed a new executive leadership team to create stronger alignment among sales, marketing, technical services, and application research. The firm had established five focus areas: household care (33% of sales in 2016), food & beverages (26%), bioenergy (17%), agriculture & feed (16%), and technical and pharma (8%). (See **Exhibit 12** for details on sales per industry.) The company had also created a central research, innovation, and supply organization spanning the value chain from discovery to market.⁵² (See **Exhibit 13** for an organizational chart.)

Competitive Environment

The company was the world's leading producer of industrial enzymes due to its first-mover advantage and proven track record of rolling out successful products.⁵³ In 2016, it accounted for 48% of the total industrial enzyme market value of DKK 25 billion. It was followed by American conglomerate DuPont (20%) and Dutch firm DSM (6%).⁵⁴ Combined, the three companies controlled about three fourths of the market, which resulted in relatively benign competitive pressures.⁵⁵

Novozymes' diverse client base over a wide spectrum of industries showed the company was a true leader and not a niche player with strengths in specific but limited areas. Analyst Natalie Koo explained that the industry hierarchy was likely to persist, mainly because of Novozymes' significant relative market share over its closest competitor.⁵⁶ Given that Novozymes' enzyme solutions were highly integrated into the manufacturing processes of its customers, switching costs were also high.⁵⁷ In addition, Novozymes' proven ability to grow faster than GDP and competitors while maintaining top-notch ROIC and return on equity (ROE) levels was a winning formula.⁵⁸

Similar to other knowledge-based industries, 21% of Novozymes' workforce (1,400 employees) and 13% of revenue were dedicated to R&D. A substantial amount of the company's intellectual property was protected by an extensive patent portfolio (7,000 patents), which safeguarded its proprietary know-how and enzyme systems from unlicensed competition for up to 20 years.⁵⁹ Most of Novozymes' products took between two and five years to go through the innovation pipeline from idea to market. For some industries, especially within food and feed products, there was a lengthy regulatory process before a product could be sold commercially. Novozymes also continuously updated its product portfolio to maintain product value and generate new patentable innovations. The innovation pipeline

contained more than 100 research projects across the business. In 2016, Novozymes had launched eight new products compared with six product launches in 2015.⁶⁰ (See **Exhibit 14** for details on the pipeline of significant growth opportunities.)

Novozymes' strategy when looking to expand into new areas was to assume either the technology risk (developing unproved technology) or the market risk (entry into new markets).⁶¹ Where the company saw an opportunity that was promising, but that entailed both technology risk and market risk, it typically searched for a partner to help develop the opportunity while sharing the risk.⁶² For example, Novozymes had teamed up with American giant Monsanto to improve crop harvests—the BioAg Alliance. Their inoculants helped plants absorb nutrients and protect against pests, disease, and weeds.⁶³ Similarly, Novozymes and DSM had developed products for animal nutrition that helped improve digestion processes and make the most of the feed, while minimizing feed costs.⁶⁴

Corporate Governance

Company Ownership

Since the 2000 demerger, Novozymes's equity had traded on the Copenhagen Stock Exchange. Its common stock consisted of two types: Class A shares and Class B shares, each with a nominal value of DKK 2 per share. All of the Class A stock was held by Novo A/S, and an A share carried 10 times as many votes as a B share. At the end of 2016, Novo A/S held 25.75% of the total common stock and, through its ownership of the A stock and a sizable stake in the B stock (26,071,400 shares), it controlled 71% of the votes.⁶⁵ Novo A/S, domiciled in Hellerup, Denmark, was wholly-owned by the Novo Nordisk Foundation. On that basis, Novozymes was included in the consolidated financial statements of the Novo Nordisk Foundation.

At year-end, Novozymes had more than 60,000 shareholders, more than 95% of whom were private shareholders in Denmark. Forty institutional investors, including Novo A/S, owned about 50% of the B shares. Around 65% of the B shares were held outside Denmark. Novozymes held 4.9% of the B stock, equivalent to 4% of the total common stock.⁶⁶ (See **Exhibit 15** for more details on the structure of the group and share price.) Nielsen observed that, "The fact that we have one long-term shareholder—the Novo Nordisk Foundation—who is very focused on sustainability, gives us an interest in advancing the sustainability agenda. It's a buffer against short-term approaches to business. It makes my life easier." COO Videbæk added, "But if we didn't deliver the results, I feel absolutely sure that our shareholders, including the Foundation, would come and say: 'This is not the way the company should be run.' We still have to be in the top tier of our industry worldwide."

A Two-Tier Governance System

In accordance with Danish law, Novozymes had a two-tier corporate governance system comprising the board of directors and the executive leadership team, with no individual being a member of both. The division of responsibilities was clearly outlined.⁶⁷

The board of directors was required to have four to eight members elected at the annual shareholders' meeting. In 2017, the board had six members. They were elected for one year at a time and could not be elected or re-elected after reaching the age of 70. Nominations were based on an evaluation of factors such as competencies, diversity, independence, and prior performance. The board of directors also included three employee-elected members, who served four-year terms.⁶⁸ (See **Exhibit 16** for biographies of the board members.)

In order to ensure the right competencies and promote diversity, the following targets were set for the composition of the board of directors. At least half of the shareholder-elected board members had to be independent as defined in the Danish recommendations on corporate governance. At least 40% of the shareholder-elected board members had to have substantial international experience from the management of large corporations or institutions headquartered outside of Denmark. One-third or more of the shareholder-elected board members had to be female, and one-third or more of the shareholder-elected board members had to be male. Only the first two targets were met in 2016.⁶⁹ The board's responsibilities were to ensure the right management and organizational structure, supervise financial, social, and environmental performance and the executive leadership team's day-to-day management of the firm, and decide the overall strategic development of the company.

The board had a chairmanship consisting of two members—the chairman and the vice chairman—that was responsible for assisting the board of directors in matters concerning the executive leadership team's day-to-day management of the firm and reporting back to the board of directors. The chairmanship was also responsible for planning and preparing meetings of the board of directors, preparing material for the nomination of candidates for election to the board of directors, and recommending the remuneration of the board of directors and the executive leadership team. The board of directors held seven meetings in 2016, with an overall attendance rate of 100%.⁷⁰

In 2017 the board of directors had established a nomination and remuneration committee, with responsibility for senior executive compensation and director nomination—duties that, to that point, had been carried out by the chairmanship. In addition, the board of directors had an audit committee that assisted the board in monitoring aspects relating to accounting, auditing, internal controls, and financial reporting.⁷¹

The six-member executive leadership team, in turn, had responsibility for the firm's daily operations. (See **Exhibit 17** for detailed biographies.) Board member Raaschou-Nielsen said, "The executive team formulates the strategy; the board debates it and improves it." Nielsen added, "As the CEO of the firm, I am not a member of the board of directors, but I will sit in on most meetings. My role is to make sure that we develop the company according to previously-agreed strategies."

Sustainability at Novozymes: What Next?

Novozymes' leaders, managers, and shareholders agreed that sustainability would remain important to the company's agenda going forward. At the board level, board members were aligned with management's approach and explicitly took into account the environmental and social performance of the firm, in addition to financial performance. The dynamics were similar among investors. Videbæk explained, "I have never been in heated discussions with investors on whether or not our sustainability focus was worthwhile. It's quite the opposite. We have discussions on whether or not we go far enough." Despite this consensus on the merits and relevance of Novozymes' sustainability focus, senior management continued to ponder the optimal speed, scale, and innovation level the company should commit to. How big a bet should the company make in looking to align with the future it believed in and anticipated—but couldn't be sure about?

Hedging its Bets?

Some team members wanted to adopt a somewhat more prudent and cautious approach. In 2016, business had been slower than expected. With sales of DKK 14.14 billion and a net profit of DKK 2.82 billion, the financial results had been satisfactory, but organic sales growth had been harder to achieve (only 1% in 2016, below the firm's initial target of 3-5%). This had been due to uncertainties stemming

from lower commodity prices, sharp changes in exchange rates, and slower growth in emerging markets. Nielsen said, "Since I have made sustainability my business, it worries me that the growth that comes from these solutions is lower than expected." For the first time in years, the company had laid off substantial numbers of staff: 198 employees in 2016, 62 of them in Denmark.⁷²

Nielsen explained, "A firm such as Novozymes that sells biological solutions that reduce the use of raw materials and energy in various processes, creates less value for its customers when energy and raw materials are cheaper. In 2016, our customers started designing their offerings based on cheaper raw materials. If we compare 2015 and 2016, the world has not changed significantly, but customer perception of what adds value has changed."⁷³

In addition, some members were concerned that governmental policy making could adversely impact the SDG agenda, and consequently the efforts carried out by Novozymes. The U.S. withdrawal from the Paris climate agreement, for example, had created anxiety. Keeping in mind that global sentiment could move in many different directions, Nielsen said, "The thing that worries me is that the transformation of our societies towards a more sustainable world is not going as fast as desired." Fordyce added, "We anticipated that the increasing focus on environmentalism would lead to more opportunities. The mistaken assumption was that the world was relatively unified. But the crisis hit and diverted people's attention. People get quite short-term when things aren't going right."

As a result, the firm had started to reallocate resources to more immediately productive uses – while waiting more patiently for some selected markets to mature. One instructive example was the temporary setback that Novozymes had experienced in the cellulosic biofuel sector.

Biofuels came in a number of forms and could be subdivided into generations. First generation biofuels, known as 'conventional biofuels', were made from feedstocks that could also be consumed as human food, whether it was sugar, starch, or vegetable oil. Over time, concerns, such as cost, inefficiency, or the impact on the food supply, arose and led to second-generation biofuels.⁷⁴ Second-generation biofuels were also produced from sustainable feedstock but these feedstocks were not normally used for human consumption. Since these feedstocks were a bit more difficult to extract, advanced conversion technologies were needed in this process, hence the label 'advanced biofuels'.⁷⁵ Second-generation technologies included, for example, cellulosic processing. Fuglsang explained:

The biofuel market in the U.S. went from nothing in the late 1990s to 14 billion gallons 14 years later. It was mostly driven by the renewable fuel standard (RFS) program – a federal program that required transportation fuel sold in the U.S. to contain a minimum volume of renewable fuels. Measures were also in place for cellulosic biofuel. Our investment in this area was opportunistic: we received significant funds from the U.S. Department of Energy to develop our technology. The investment in cellulosic biofuel was considered with significant higher risk but also with potential significant rewards, as enzyme volumes required for this industry were massive. Around 2008, we started investing significant amounts of money in this sector. Cellulosic biofuels were our largest research endeavor at the time. 150 researchers were working on this full time. A large part of our share price was associated to this opportunity. It could have transformed Novozymes and doubled the size of the firm in a short period of time. But it didn't happen just quite like that: oil prices plummeted and eroded the climate for risky investments in biomass. Investments stalled and cellulosic biofuels became less attractive—at least temporarily. Biomass policies around the world started changing. Today we supply enzymes to advanced bio-refineries, but the market and processing technologies didn't advance at the pace that we had envisaged.

In such circumstances, according to Fordyce, resource reallocation became a necessity:

It's not like we are disengaging, but we think that the allocation of resources needs to shift to reflect the fact that those efforts for the time being are less impactful than we expected and we need to produce higher growth rates. Therefore we'll shift more of our resources towards emerging market commercial opportunities and a subset of higher-impact R&D projects, and less towards sustainability and government engagement. We want the world to go in a certain direction but we can't pull it there by ourselves.

Doubling down?

On the other hand, some members wanted to embrace a more radical outside-in approach in R&D, placing still more high-risk, high-return bets. Pedersen, the self-described "activist," advocated going all-in, "Obviously that will require a new way of thinking. It is not business as usual. It's more risky and uncomfortable. It's a new way to drive business compared to what we have done historically."

Pedersen proposed redoubled efforts to close performance gaps in operations and supply chain, revamp environmental, social, and governance (ESG) reporting, and further developing and applying the Global Goal assessments. He also advocated new initiatives to inform and engage employees in sustainability, a stronger alignment of public affairs and sustainability work at corporate and regional level, and the mandatory inclusion of sustainability insights in strategy development for sales, marketing, and business development. And he wanted to deliver on "Partnering for Impact" projects in animal feed, bioenergy, palm oil, and water, to professionalize the approach to non-commercial partnering, and to follow a more radical outside-in approach to innovation and R&D, starting with societal needs and working backwards from there. Pedersen commented, "That's mostly a question of ambition and speed because we are already doing many of the right things."

Novozymes' success would also have a lot to do with the politics of climate change, environmentalism, and unpredictable fluctuations in public opinion. Fordyce explained:

Let's take first generation ethanol: there are people in the world that feel that it's the worst thing that ever happened, because we are diverting edible commodities into fuel. So there is a controversy around these areas because of the way society thinks of them. We take that in; we do our own assessment; we don't have to agree with everybody, so we establish our own view. But it's sensible to take these things more seriously. Because at worst, if you don't, you might get caught on the wrong side of a particular issue that consumers got alerted to. At best, you can proactively position yourself as somebody who cares about this.

Accordingly, despite being viewed by many as a "pure play technology company"—working behind the scenes, with no direct connection to average consumers—Novozymes became very attentive to public attitudes that would likely influence the commercial success of its solutions. In fact, the company sought to actively influence the ecosystems in which it operated around the world, including by weighing in on regulatory debates with the goal of nudging them toward being more hospitable to Novozymes' solutions. Nielsen said, "Technology can make us meet the 17 SDGs. But we need the right political framework for this."

For example, Novozymes had successfully worked with the Chinese Cleaning Industry Association (CCIA)—the most important stakeholder of the Chinese detergent industry. The goal of partnering with the CCIA was to work toward higher standards in limiting the permissible negative environmental impacts of laundry detergents. With its high GDP growth and a fast-growing middle

class with increasing spending power, the Chinese market was extremely attractive to Novozymes. But less stringent sustainability standards had slowed down enzyme penetration in the country. Fuglsang said, "We thought: 'Why don't we work with the Chinese government to start legislating on what the detergent standards should be in the country? We should try to bring facts in the system to guide the decision process. We are on to something'." Pedersen added, "By working with a partner that was well connected to the government, we created a momentum. We changed detergent regulations. China benefited from that as a society. We benefited from that in our business. I think that we should work this way in the future: identify the opportunities for improvements at a societal level, find out who the partners are, who has enough power to accelerate these developments, so that society and Novozymes could both benefit."

According to this view, partnering would be critical for Novozymes, since its ability to singlehandedly influence the ecosystems where it operated was quite limited. Fuglsang explained, "Hopefully, the political system will reform itself and embrace what is happening and the changes the world is facing. In the meantime, we are just a mid-size European company and what we can do on our own is limited. By teaming up with other large companies or organizations, we can get a multiplier effect, especially in emerging markets which seek to 'leapfrog' developments."

Pedersen aspired to go still one step further. He believed that Novozymes should engage all the stakeholders affected by social and environmental problems, and help move them towards collaborative action. The aim was to transform the economy and society in ways that went beyond the reach of Novozymes as a single organization. Put bluntly, Pedersen wanted to "change the system." He said, "We need to think, work, act in a systemic way instead of business to business."

What Next?

Stepping out of the elevator and into the Novozymes offices, Nielsen wondered about the best path forward for Novozymes. Was more caution called for, in light of all the contextual uncertainties that would surely impact the firm's fortunes, for better or worse? Or should the company still be bolder than it had been to date—and perhaps even ratchet up its nascent efforts to actively shape those contexts?

Novozymes had been engaged in the SDG process since its very beginning, and had successfully embraced sustainability as a key driver of innovation in recent years. Nielsen said, "It's not like the 17 UN goals are a religion but they express in a good way some of the major challenges that the planet will be facing. We should think about long-term value creation. We shouldn't get caught in short-term dramas." Nielsen thought back to his recurring conversations with Pedersen on the evolution of the company and the way forward. He recalled Pedersen saying that he wanted to accelerate the sustainability agenda—convinced that the company could simultaneously do good and do well. Nielsen was aware, though, that "at the same time, the short term has to be bright; this is how we maintain our license to operate for the longer term."

As Novozymes' CEO, Nielsen knew that he needed to propose a promising path forward to the board, and then to rally investors and other key stakeholders behind the company's strategy. Sitting down at his desk, he anticipated the lively discussion in the Novozymes boardroom that lay ahead.

Exhibit 1 Novozymes Sustainability Journey (as of 2017)

Source: Company documents.

Exhibit 2 The United Nations 2030 Sustainable Development Goals (as of 2017)Source: United Nations, "Sustainable Development Goals," United Nations Website, <http://www.un.org/sustainabledevelopment/sustainable-development-goals/>, accessed August 2017.

Exhibit 3 Novozymes' Impact Categories (as of 2017)

Source: Company documents.

Exhibit 4 Updated Operational Targets (as of 2017)

Environment	People
- Energy efficiency - Water efficiency - Reduction in CO₂ intensity - Renewable Energy - Supplier Program - Gold Class RobecoSAM rating - Carbon disclosure project A list	- Occupational accidents - Employee absence - Employee satisfaction and motivation - Employee opportunities for development - Women in senior management

Source: Company documents.

Exhibit 5 Illustration of Process to Identify Global Goal High Impact Opportunities in Novozymes Innovation Pipeline (as of 2017)

Source: Company documents.

Exhibit 6 Consolidated Environmental Data (2015–2016)

		2015	2016
Climate change			
Estimated CO ₂ reductions*	Million tons	60	69
Greenhouse gas emissions	1,000 tons CO ₂ eqv.	408	413
CO ₂ intensity	%	17	16
Energy			
Energy consumption	1,000 GJ	4,148	4,380
Renewable Energy	%	24	24
Energy efficiency	%	15	10
Water			
Water consumption	1,000 m ³	6,965	7,225
Volume of wastewater	1,000 m ³	4,917	5,392
Water efficiency	%	9	6
Waste			
Solid waste	1,000 tons	13	13
Recycling rate	%	50	44
Biomass volume	1,000 tons	517	550
Environmental compliance, etc.			
Breaches of regulatory limits	Number	27	37
Neighbor complaints	Number	11	9

Source: Company documents.

Note: * From customers' application of Novozymes products in their products or processes.

Exhibit 7 Consolidated Social and Governance Data (2015-2016)

		2015	2016
Labor practices and human rights			
Employee, total	Number	6,485	6,441
Women	%	37.5	37.7
Men	%	62.5	62.3
Rate of absence	%	2.0	2.0
Employees promoted who are women	%	41	36
"Satisfaction & Motivation" score in employee survey	Number	77	80
"Opportunities for professional and personal development" score in employee survey	Number	80	79
Occupational health and safety			
Fatalities	Number	0	0
Frequency of occupational accidents	Per million working hours	2.5	2.2
Frequency of occupational diseases	Per million working hours	1.2	0.7
Innovation			
New products	Number	6	8
Active patent families	Number	1,164	1,123
Business ethics			
Completion of business integrity training for employees*	%	98	99
Breaches of competition law	Number	0	0
Investigates fraud cases*	Number	25	44
Corporate citizenship			
Estimated number of learners reached	Number	25,000	106,000
Customer satisfaction measurement			
Net Promoter Score (NPS)	Number (-100 to +100)	N/A	45

Source: Company documents.

Exhibit 8 Novozymes' Long-Term Targets: Achievements (2016)

Targets	Achievements in 2016
Reach 6 billion people with our biological solutions	- We reached approximately 5 billion consumers with more than one of our solutions on a weekly basis- approximately 100 million more than last year. - This growth was driven by the global increase in textile products and by laundry products in China - Furthermore, we have refined how we calculate delivery of this target. What's next? The growth potential of the REACH target is greatest in India, China, and Africa, with the laundry and animal feed industries being the key growth drivers.
Educate 1 million people about the potential of biology	- Novozymes is dedicated to encouraging local education with global impact by driving EDUCATE activities out of our largest regions: China, India, Brazil, North America and Europe, Middle East & Africa (EMEA) . - In 2016, Novozymes educated more than 100,000 learners through regional partnerships. This brings our total to more than 130,000 learners since 2015. What's next? In the coming year, we will continue to scale up our most successful programs in all regions. We will also develop new partnerships with relevant educational organizations.
Catalyze 5 global partnerships for change	- Forming new strategic partnerships is not easy, and developing strong, impactful partnerships that have lasting and transformational impact on business takes time and effort - In 2016, we made headway on forming promising partnerships for change. One of these partnerships was with DONG Energy. Novozymes will supply enzymes to the world's first energy plant turning household waste into biogas, electricity and fuel. Novozymes and DONG Energy have also agreed to further develop the enzymes for the technology. What's next? In 2017, we will dedicate additional organizational resources to help accelerate new promising partnerships as well as focus on further enhancing the partnering culture across Novozymes.
Deliver 10 transformative innovations	- Acceleron® B-300 SAT, our new-generation corn seed treatment developed with Monsanto, represents a step-change improvement in crop yield and resilience. What's next? We have a number of other promising transformational innovations in our pipeline that are expected to launch in 2017 and contribute to this target.
Save 100 million tons of CO₂	- Based on LCAs – from raw material extraction, through production and use, to final disposal – we estimate that our solutions saved customers a total of 69 million tons of CO₂ in 2016. - The main drivers of the additional savings compared with 2015 (a total of 60 million tons) were our household care, animal health & nutrition and textile products. - Fuel ethanol is one of the industries with immense potential to contribute to achievement of the SAVE target. However, 2016 saw a slight decline in this contribution due to more or less flat sales in terms of product volume What's next? Delivery on the CO₂ savings target is closely connected to the volume of various enzymes brought to market. On top of the volume growth of the existing product portfolio, we continue to explore other opportunities to

Targets	Achievements in 2016
Enable Novozymes' employees to develop	increase our CO₂ savings, for example by increasing sales of products with particularly positive CO₂-saving profiles, either from our existing portfolio or by further accelerating specific innovations in our pipeline. - 90% of all Novozymes employees have Individual Development Plans with development targets and actions, exceeding our target of 80%. - Our dedication to employee development is measured through our annual employee survey. With a score of 79 in 2016, meeting our target of 75, our employees agree that development is a priority at Novozymes. - More than 2,600 employees worldwide participated in Development Week to enhance their personal and professional development skills. - Regional leadership pipelines were strengthened through targeted talent development initiatives. In China, for example, 62% of identified and developed talents were promoted, had their role expanded or were assigned new roles. In North America, 50% of identified and developed talents were promoted. What's next? Over the coming years, we will work on the following focus areas to "Enable Novozymes' employees to develop:" - Unfolding the potential of talents across our global organization with a special focus on building capabilities and talent in our high-growth markets and high-investment business areas. - Developing leaders to be capable of leading a multigenerational, multicultural workforce in a changing business environment. - Promoting more agile working structures and building the skills and mindset required to embrace digitalization

Source: Company documents.

Exhibit 9 Roles of Enzymes and Microorganisms (as of 2017)

What are Ezymes?	What do Enzymes do?
Enzymes are proteins that act as catalysts and thereby help complex reactions occur everywhere in life. This is nature's own way of kick-starting biological processes. When one substance needs to be transformed into another, nature uses the catalyzing enzymes to speed up and control the process. With Novozymes' expertise, enzymes can excel on an industrial scale by optimizing output and accelerating processes, while saving water, energy, and raw materials.	- Animal Feed: maximize nutrition by enhancing digestive systems - Baking: improve softness, freshness, and dough strength - Biofuels: produce sustainable energy from crops or waste - Brewing: improve filtration and flavor of beer and ensure raw material optimization - Detergents: remove tough stains and enable low-temperature wash - Health and nutrition: make foods safer by removing carcinogenic and allergenic ingredients - Juice: increase yields and improve clarity of juice - Leather: prepare, degrease, and tan leather more efficiently - Oils and fats: harden fats without unhealthy trans-fatty acids - Pulp and paper: reduce water, energy, and chemical consumption - Pharmaceuticals: improve yields and reduce costs of drug production - Starch: produce sugar syrups more simply and efficiently - Textile: save time, water, energy, and chemicals in textile production.
What are Microorganisms?	What do Microorganisms do?
Microorganisms are living, single-celled organisms such as fungi or bacteria. Like enzymes, Novozymes' microorganisms have natural properties that influence processes and reactions. The industrial applications for microorganisms are countless. For example, microbes expand the farmer's toolkit to increase yield and protect crops. Microbes produce enzymes and are found everywhere, some come from volcanoes and Artic lakes and are particularly robust in hot and cold environments.	- Biological solutions for agriculture (BioAg): improve crop yield and health - Poultry health and nutrition: improve animal health, growth, and feed utilization - Aquaculture: support improved growth, survival, and disease resistance - Biogas: work as processing aids - Wastewater treatment: make wastewater treatment facilities more efficient.

Source: Company documents.

Exhibit 10 Selected Novozymes Financials (2015–2016, DKK)**a) Balance Sheet**

Balance sheet (DKK million)	2015	2016
Non-current assets	11,643	12,266
Total assets	17,791	18,659
Shareholders' equity	11,593	11,745
Non-current liabilities	2,898	3,352
Current liabilities	3,300	3,562
Invested capital	11,891	12,584
Net interest-bearing debt	437	990

b) Income Statement

Income Statement (DKK million)	2015	2016
Revenue	14,002	14,142
Research and development costs	(1,896)	(1,865)
EBITDA	5,011	4,960
Operating profit / EBIT	3,884	3,946
Financial items, net	(257)	(34)
Profit before tax	3,621	3,881
Net profit	2,825	3,050

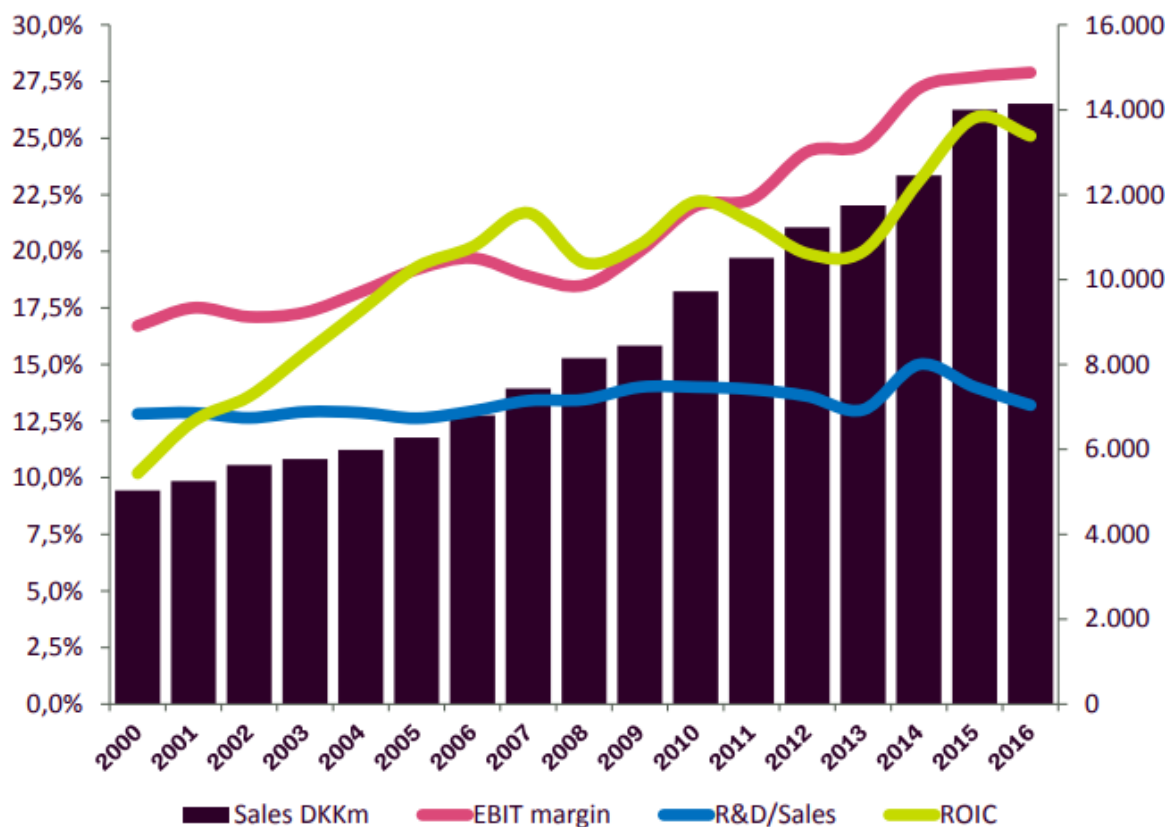
c) Cash Flow Statement

Cash Flows (DKK million)	2015	2016
Cash flow from operating activities	3,339	3,840
Cash flow from investing activities, net	(1,257)	(1,349)
Of which investments in property, plant and equipment, net	(968)	(1,076)
Free cash flow before acquisitions	2,324	2,652
Free cash flow	2,082	2,491
Cash flow from financing activities	(3,681)	(2,484)
Net cash flow	(1,599)	7

d) Key Ratios

Key ratios	2015	2016
Research and development costs as a percentage of revenue (%)	13.5	13.2
EBITDA margin (%)	35.8	35.1
Operating profit margin (%)	27.7	27.9
Effective tax rate (%)	22.0	21.4
Equity ratio (%)	65.2	62.9
Return on equity (%)	24.7	26.1
Return on invested capital (ROIC), including goodwill (%)	25.9	25.1
WACC after tax (%)	5.3	6.5
Earnings per share (EPS), diluted (DKK)	9.12	10.06
Proposed dividend per share (DKK)	3.50	4.00

Source: Company documents.

Exhibit 11 Novozymes' Historical Performance (2000–2016)

16 years of innovation-driven returns

Strong returns since 2000:

- 2.8x sales
- 4.7x EBIT
- 6.3x net profit
- DKK 36 billion in operating cash flows

Innovation is the recipe:

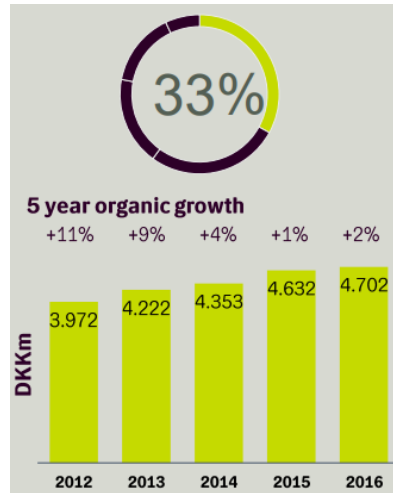
- DKK 20 billion invested in R&D
 - +100 new solutions marketed
 - +5.5% in gross margins expansion through productivity improvements
-

Source: Company documents.

Exhibit 12 Novozymes Market Overview: Sales by Industry (2012-2016)

Household Care

Percentage of 2016 Sales:



Applications:

laundry and dishwasher detergents

Novozymes market share:

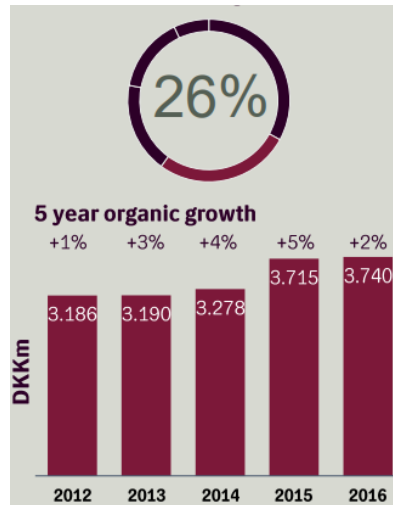
>60%

Competition:

Dupont

Food and Beverages

Percentage of 2016 Sales:



Applications:

Baking, starch conversion, brewing, beverage alcohol, healthy concepts

Novozymes market share:

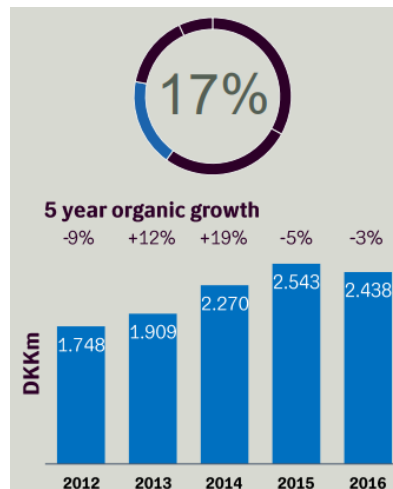
30%-40%

Competition:

DSM, Dupont, regional

Bioenergy

Percentage of 2016 Sales:



Applications:

Corn ethanol, biomass conversion, mainly a U.S. business today (85%)

Novozymes market share:

50%

Competition:

Dupont, regionals, and other technologies

Agriculture and Feed

Percentage of 2016 Sales:

Applications:

Enzymes for nutritive enhancement of feed for pigs and chicken.
Microbes for crop yield enhancement. Probiotics.

Novozymes market share:

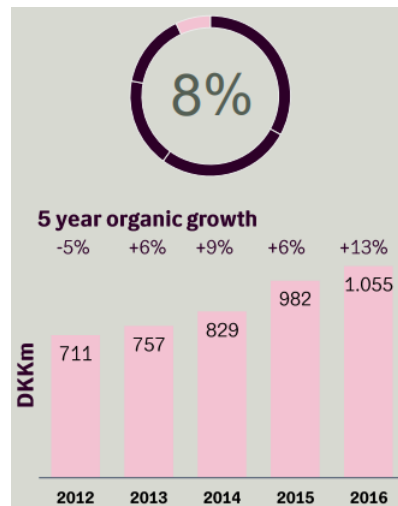
25%-30% (feed enzymes)
Leading producer of inoculants (BioAg)

Competition:

Dupont, AB Vista, regional (feed enzymes), BASF, Bayer, regional (BioAg)

Technical and Pharma

Percentage of 2016 Sales:

Applications:

Textiles, leather, pulp and paper, albumin, biocatalysis

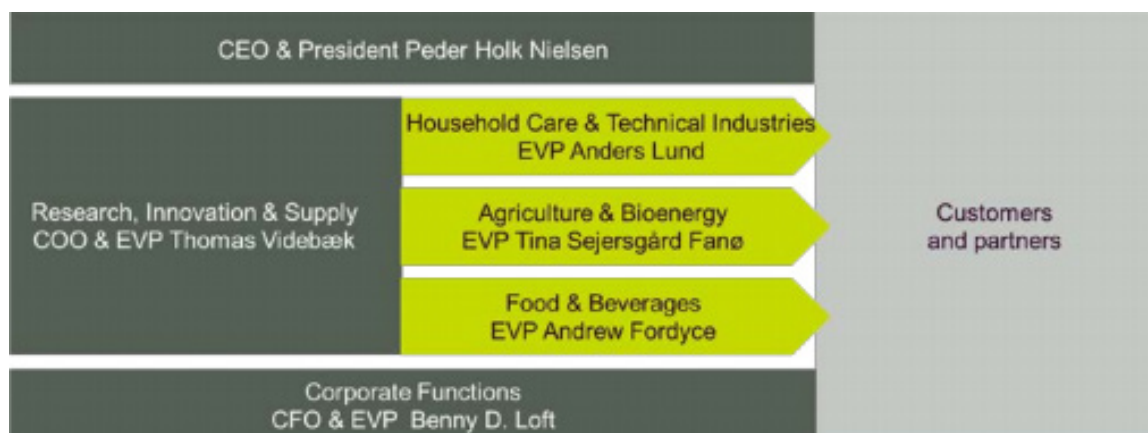
Novozymes market share:

50% (enzymes only)

Competition:

Dupont, regional

Source: Company documents.

Exhibit 13 Novozymes' New Organizational Structure (since February 2016)

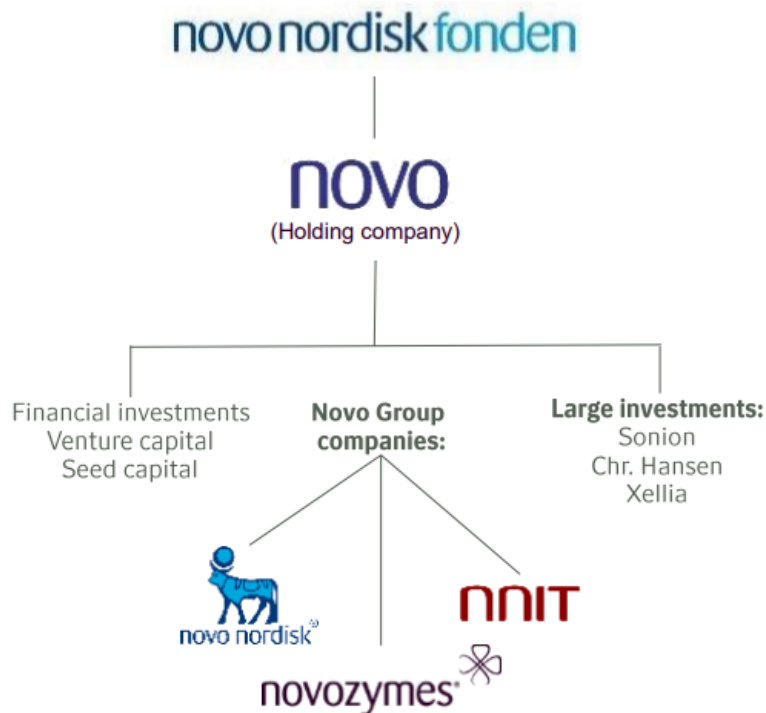
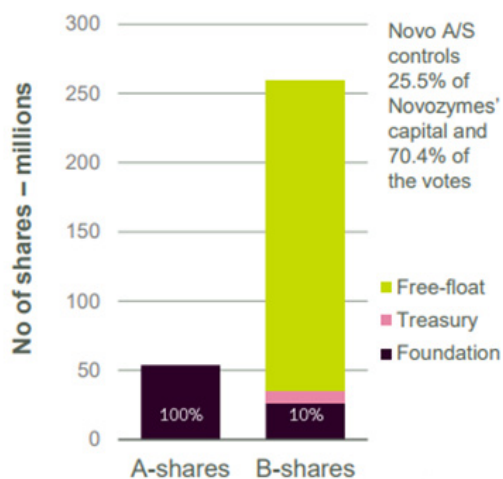
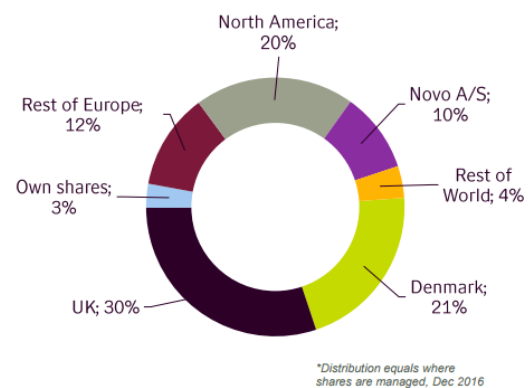
Source: Company documents.

Exhibit 14 Pipeline of Significant Growth Opportunities (as of 2016)

Area	Innovation	Feasibility	Discovery	Development	Launch	Commercial
Household Care	Hygiene solutions	✓	✓	●		
Household Care	Tailored emerging markets solutions	✓	✓	●		
Food & Beverages	Vegetable oil processing	✓	✓	●		
Food & Beverages	Grain milling	✓	✓	→	●	
Agriculture & Feed	Natural growth promotion	✓	✓	✓	●	
Agriculture & Feed	Enhanced corn inoculant	✓	✓	✓	✓	→
Agriculture & Feed	New transformative BioAg solutions	✓	●			
Bioenergy	Biomass conversion	✓	✓	✓	●	

Arrows denote progress to next phase over the past three months

Source: Company documents.

Exhibit 15 Structure of the Group**a) Ownership (as of 2017)****b) Shareholder Distribution: A-Shares and B-shares (as of 2017)****Shareholder distribution of B common stock***

Novo A/S was the only major stockholder holding more than 5% of Novozymes common stock Dec 31 2016

c) Novozymes' Share Price (April 2016–April 2017, DKK)



-26% in 2016
 +12% in 2017
 Current market value: DKK 84 billion
 The stock has grown by a factor of 9 since the IPO in 2000
 Total return over last 5 years: 60%

Source: Company documents.

Exhibit 16 Novozymes' Board of Directors (as of 2017)

Board Member	Biography
Agnete Raaschou-Nielsen 	Born 1957. <i>Danish</i>. Chairman of the Board of Directors since 2017. Member of the Nomination and Remuneration Committee. Member of the Board since 2011. Elected for one year at a time. Board positions Chairman: Arkil Holding A/S Brdr. Hartmann A/S Danske Invest, three other UCITS funds and two AIF funds Vice Chairman: Dalhoff Larsen & Horneman A/S Solar A/S Member: Aktieselskabet Schouw & Co. Danske Invest Management A/S Member of the Audit Committee: Aktieselskabet Schouw & Co. Solar A/S Special competencies Expertise within business development and acquisitions, macroeconomics and intellectual property rights. Experience in finance and accounting matters.
Jørgen Buhl Rasmussen 	Born 1955. <i>Danish</i>. Vice Chairman of the Board. Adjunct Professor at Copenhagen Business School. Member of the Board since 2011. Member of the Audit Committee. Member of the Nomination and the Remuneration Committee. Elected for one year at a time. Board positions Chairman: F. Uhrenholt A/S Member: IFC Europe A/S Human Practice Foundation Advisory Board of Axcel Special competencies International business and management experience, specifically within sales, marketing, branding and acquisitions. Experience in finance and accounting matters.
Lars Green 	Born 1967. <i>Danish</i>. Vice Chairman of the Board. Senior Vice President, Finance & Operations, Novo Nordisk Inc. (US). Member of the Board since 2014. Elected for one year at a time. <i>This board member is not regarded as independent in the sense of the definition in the Danish Recommendations on Corporate Governance that apply to Danish listed companies.</i> Special competencies In-depth knowledge of the Novo Group's business, international experience from managing global biotech and biopharma companies, and financial and accounting expertise.

Board Member	Biography	
Kasim Kutay 	Born 1965. <i>British</i>. Member of the Board. Chief executive officer of Novo Holdings A/S. Elected for one year at a time. <i>This board member is not regarded as independent in the sense of the definition in the Danish Recommendations on Corporate Governance that apply to Danish listed companies.</i> Board position: Member: ConvaTec plc, UK Special competencies Extensive experience as financial advisor to the pharmaceutical, biotechnology and medical device industries.	
Kim Stratton 	Born 1963. <i>Australian</i>. Member of the Board. Member of the executive management of Shire, a global biotech company, where she is responsible for all commercial activities outside the US. Member of the Nomination and the Remuneration Committee. Elected for one year at a time. Special competencies Broad international commercial experience within sales and marketing, emerging markets, innovation pipeline management and sustainability.	
Mathias Uhlén 	Born 1954. <i>Swedish</i>. Professor at the Royal Institute of Technology (KTH) in Sweden and the Technical University of Denmark (DTU). Member of the Board since 2007. Elected for one year at a time. Board positions Chairman: Atlas Antibodies AB Antibodypedia AB Vice Chairman: Atlas Antibodies AB Member: Alligator AB Bure Equity AB Woodheads AB Special competencies Broad experience in research and biotechnology.	
Anders Hentze Knudsen 	Lars Bo Køppler 	Lena Bech Holskov 
Born 1959. <i>Danish</i>. Senior Operator. Employee representative. Member of the Board since 2013. Elected for four years at a time.	Born 1952. <i>Danish</i>. Technician. Employee representative. Member of the Board since 2010. Elected for four years at a time.	Born 1967. <i>Danish</i>. Safety Adviser. Employee representative. Member of the Board since 2013. Elected for four years at a time.

Source: Company documents.

Exhibit 17 Novozymes' Executive Leadership Team (as of 2017)

Executive Member	Biographies
Peder Holk Nielsen 	Born 1956. President & CEO. Board positions Member: Hempel A/S LEO Pharma A/S Education: Holds a Ph.D. and M.Sc. in Chemical Engineering from the Technical University of Denmark (DTU) and a B.Com. in International Business Management from Copenhagen Business School. Special competencies Novozymes' CEO since 2013. Peder focuses on developing the organization and processes to effectively turn market insights into product ideas and solutions that excite Novozymes' customers. With his background in engineering and business management, Peder drives an agenda that couples market insights and research capabilities to deliver innovation and growth.
Tina Sejersgård Fanø 	Born 1969. Executive Vice President, Agriculture & Bioenergy. Board positions Member: DLF Seeds & Science Professional Packaging Systems Education: Holds an M.Sc. in Chemical Engineering from the Technical University of Denmark. Special competencies Tina is responsible for application research, technical service, sales and marketing in the Agriculture & Bioenergy division. Tina has significant experience in developing and managing global partnerships and has been instrumental in negotiating several major business deals for Novozymes over the years.
Andrew Fordyce 	Born 1963. Executive Vice President, Food & Beverages. Education: Holds a Ph.D. in Chemical Engineering from the University of Texas at Austin, US. Special competencies Andy is responsible for application research, technical service, sales, and marketing in the Food & Beverages division. Andy's career has moved from pure engineering to a strong focus on value generation for customers. Previous responsibilities include global sales and marketing, strategic account management, and technical service strategy.

Executive Member	Biographies
Benny D. Loft 	Born 1965. CFO & Executive Vice President, Corporate Functions. Board positions Member: DONG Energy A/S New Xellia Group A/S Chairman of the Audit Committee: DONG Energy A/S New Xellia Group A/S Education: Holds an M.Sc. in accounting, tax and auditing from Copenhagen Business School. State-authorized Public Accountant. Special competencies Benny leads Corporate Functions, which covers finance, investor relations, legal, IT, human resources, sourcing, global business services, facility management, and communications. Benny combines deep financial experience and acumen with extensive operational knowledge of the company's core areas.
Anders Lund 	Born 1973. Executive Vice President, Household Care & Technical. Education: Holds an M.Sc. in Economics from Aarhus University, Denmark. Special competencies Anders is responsible for application research, technical service, sales and marketing in the Household Care & Technical division. Anders has a strong commercial and strategic background as well as extensive experience of building and maintaining global customer relationships.
Thomas Videbæk 	Born 1960. COO & Executive Vice President, Research, Innovation & Supply. Board positions Vice Chairman: Albumedix A/S Member: Evolva SA Education: Holds a Ph.D. and an M.Sc. in Chemical Engineering from the Technical University of Denmark (DTU) and a B.Com. in International Business from Copenhagen Business School. Special competencies Thomas is responsible for Novozymes' Research, Innovation & Supply unit. The unit has core research at its center and focuses on developing new biological solutions and production optimization – from discovery to large-scale manufacturing. The unit also ensures the supply and quality of our products and incubation of new platforms. Thomas has been a central driver of business ventures outside Novozymes' established areas for several years. He also has broad knowledge of sales and customer solutions and supply chain operations.

Source: Company documents.

Endnotes

- ¹ Novozymes, 2016 Annual Report, (Bagsværd: Novozymes, 2016), <https://report2016.novozymes.com/>, p. 4, accessed August 2017.
- ² Novozymes, "Novozymes' History: Our Heritage," Novozymes Website, <https://www.novozymes.com/en/about-us/history>, accessed November 2017.
- ³ World Business Council for Sustainable Development, "SDGs as high impact innovation opportunity," WBCSD Website, April 5, 2016, <http://wbcspdpublications.org/sdgs-as-high-impact-innovation-opportunity/>, accessed November 2017.
- ⁴ Novozymes, 2016 Annual Report, p. 25.
- ⁵ Novozymes, "Fact Sheet," Novozymes Website, https://www.novozymes.com/en/-/media/Novozymes/en/news/fact-sheet/Documents/Novozymes_factsheet-2016-EN.pdf?la=en, accessed August 2017.
- ⁶ Novo Nordisk "The History of Novo Nordisk," Novo Nordisk Website, October 2014, <http://www.novonordisk.co.uk/content/dam/UK/AFFILIATE/www-novonordisk-co-uk/Home/Commons/Documents/The%20History%20of%20Novo%20Nordisk.pdf>, accessed August 2017.
- ⁷ Novo Nordisk Fonden, "History of the Novo Nordisk Foundation," <http://novonordiskfonden.dk/en/content/history-novo-nordisk-foundation>, accessed August 2017.
- ⁸ Novo Nordisk "The History of Novo Nordisk."
- ⁹ Novo Nordisk Fonden, "History of the Novo Nordisk Foundation."
- ¹⁰ Company Histories, "Novo Nordisk A/S," Company Histories Website, <http://www.company-histories.com/Novo-Nordisk-AS-Company-History.html>, accessed August 2017.
- ¹¹ Novo Nordisk, "Novo Nordisk History," Novo Nordisk Website, https://www.novonordisk.com/content/dam/Denmark/HQ/aboutus/documents/HistoryBook_UK.pdf, p. 42, accessed August 2017.
- ¹² David J. Bennett, Richard C. Jennings, *Successful Science Communication: Telling It Like It Is*, (Cambridge: Cambridge University Press, 2011), p. 295, accessed August 2017.
- ¹³ Ibid., p. 46.
- ¹⁴ Novo Nordisk, 2000 Environmental and Social Report (Bagsværd: Novo Nordisk, 2000), p. 2, https://www.novonordisk.com/content/dam/Denmark/HQ/Sustainability/documents/2000_NN%20Environmental%20and%20Social%20Report%202000.pdf, accessed August 2017.
- ¹⁵ Novo Nordisk "The History of Novo Nordisk."
- ¹⁶ Nick Aster, "How Novozymes Integrates Sustainability into the Business," December 9, 2013, post on blog "Triple Pundit," <http://www.triplepundit.com/2013/12/novozymes-sustainability/>, accessed August 2017.
- ¹⁷ Robert G. Eccles, Michael P. Krzus, *The Integrated Reporting Movement: Meaning, Momentum, Motives, and Materiality*, (New Jersey: John Wiley & Sons, 2015), p. 34, accessed August 2017.
- ¹⁸ Novozymes, 2002 Annual Report, (Bagsværd: Novozymes, 2002), p. 5, http://www.zonebourse.com/NOVOZYMES-447531/pdf/8355/NOVOZYMES_Rapport-annuel.pdf, accessed August 2017.
- ¹⁹ Vijay Kanal, "Five Stages on the Sustainability Journey," *GreenBiz*, December 13, 2010, <https://www.greenbiz.com/blog/2010/12/13/five-stages-sustainability-journey>, accessed August 2017.
- ²⁰ Ibid.
- ²¹ Novozymes, "Rising raw material prices and Novozymes," Novozymes Website, 2008, http://report2007.novozymes.com/Menu/Reporting%2BUupdates/Q2%2B2008/Q2%2Btheme_%2BRising%2Braw%2Bmaterial%2Bprices%2Band%2BNovozymes.html, accessed August 2017.
- ²² Jenny Davis-Peccoud, Paul Stone, Clare Tovey, "Achieving Breakthrough Results in Sustainability," Bain & Company, November 18, 2016, <http://www.bain.com/publications/articles/achieving-breakthrough-results-in-sustainability.aspx>, accessed August 2017.

²³ Nick Aster, "How Novozymes Integrates Sustainability into the Business."

²⁴ Sustainable Development Goals Fund, "From MDGs to SDGs," SDGF Website, <http://www.sdgsfund.org/mdgs-sdgs>, accessed September 2017.

²⁵ United Nations Sustainable Development Platform, "Open Working Group on Sustainable Development Goals," United Nations Website, <https://sustainabledevelopment.un.org/owg.html>, accessed August 2017.

²⁶ Ibid.

²⁷ Ibid.

²⁸ Novozymes, 2016 Annual Report, p. 33.

²⁹ Ibid., p. 39.

³⁰ Jenny Davis-Peccoud, Paul Stone, Clare Tovey, "Achieving Breakthrough Results in Sustainability."

³¹ Novozymes, "Sustainability Governance," Novozymes Website, <http://www.novozymes.com/en/sustainability/novozymes-sustainability-approach/sustainability-governance>, accessed August 2017.

³² Novozymes, 2016 Annual Report, p. 3.

³³ Ibid., p. 3.

³⁴ Ibid., p. 15.

³⁵ Ibid., p. 134.

³⁶ Ibid., p. 74.

³⁷ Ibid., p. 136.

³⁸ Ibid., p. 74.

³⁹ Ibid., p. 76.

⁴⁰ Ibid., p. 76.

⁴¹ Ibid., p. 76.

⁴² Ibid., p. 41.

⁴³ Ibid., p. 15.

⁴⁴ Novozymes, "Novozymes honored by the UN," Novozymes Press Release, June 22, 2016, <https://www.novozymes.com/en/news/news-archive/2016/06/un-honors-novozymes-as-sdg-local-pioneer>, accessed October 2017.

⁴⁵ Novozymes, Roadshow Presentation, 1H 2017, Novozymes Website, https://s21.q4cdn.com/655485906/files/doc_financials/Quarterly/Q2_2017/2017_Q2_Novozymes_Roadshow-presentation_FINAL.pdf, accessed November 2017.

⁴⁶ Umar Riaz, "Novozymes offers efficient and reliable enzyme solutions for the textile industry," Pakistan Textile Journal, <http://www.ptj.com.pk/Web%202004/04-2004/interview.html>, accessed November 2017.

⁴⁷ Krishna G. Palepu, Karol Misztal, "Novozymes: Cracking the emerging markets code," Harvard Business School, Case Study No. 112-084, April, 4, 2012, accessed via Harvard Business School Publishing in August 2017.

⁴⁸ Novozymes, 2016 Annual Report, p. 6.

⁴⁹ "Novozymes' dominance of the oligopolistic industrial enzyme market results in a wide moat," Morningstar Analyst Report, January 20, 2017, accessed via Thomson One, August 2017.

⁵⁰ Ibid.

⁵¹ Novozymes, 2016 Annual Report, p. 16.

⁵² Novozymes, "Changes in organizational structure and Executive Leadership Team," Press Release, February 8, 2016, https://www.novozymes.com/en/-/media/Novozymes/en/investor/news-and-announcements/Documents/2016_3_Novozymes-changes-to-organization-and-leadership-team_FINAL_EN.pdf?la=en, accessed August 2017.

⁵³ Natalie Koo, "Enzyme Leader Novozymes: Biotech Without Biotech Risk," Seeking Alpha, June 20, 2016, <https://seekingalpha.com/article/3983112-enzyme-leader-novozymes-biotech-without-biotech-risk?page=2>, accessed August 2017.

⁵⁴ "Novozymes A/S (NZYM B) - Financial and Strategic SWOT Analysis Review," Global Data Analyst Report, April 2017, p. 20, accessed via Thomson One, August 2017.

⁵⁵ "Novozymes' dominance of the oligopolistic industrial enzyme market results in a wide moat," Morningstar Analyst Report.

⁵⁶ Natalie Koo, "Enzyme Leader Novozymes: Biotech Without Biotech Risk."

⁵⁷ "Novozymes' dominance of the oligopolistic industrial enzyme market results in a wide moat," Morningstar Analyst Report.

⁵⁸ Natalie Koo, "Enzyme Leader Novozymes: Biotech Without Biotech Risk."

⁵⁹ "Novozymes' dominance of the oligopolistic industrial enzyme market results in a wide moat," Morningstar Analyst Report.

⁶⁰ Novozymes, 2016 Annual Report, p. 32.

⁶¹ Novozymes, FY2016 Roadshow Presentation, Novozymes Website, p. 15, <https://investors.novozymes.com/investors/events-and-presentations/event-details/2017/FY-2016-roadshow-03-10-2017/default.aspx>, accessed September 2017.

⁶² Ibid.

⁶³ Novozymes, "The BioAg Alliance advances new microbial solutions for agriculture," Novozymes Website, January 5, 2017, <https://www.novozymes.com/en/news/news-archive/2017/01/bioag-alliance-pipeline-2017>, accessed September 2017.

⁶⁴ DSM, "The feed enzymes alliance," DSM Website, https://www.dsm.com/markets/anh/en_US/products/products-feedenzymes/products_feed_enzymes_alliance.html, accessed September 2017.

⁶⁵ Novozymes, "Ownership," Novozymes Website, <https://investors.novozymes.com/investors/stock-information/stock-information/default.aspx>, accessed August 2017.

⁶⁶ Ibid.

⁶⁷ Novozymes, "Corporate Governance," Novozymes Website, <https://investors.novozymes.com/investors/corporate-governance/board-of-directors/default.aspx>, accessed August 2017.

⁶⁸ Ibid.

⁶⁹ Ibid.

⁷⁰ Ibid.

⁷¹ Ibid.

⁷² Novozymes, 2016 Annual Report, p. 16.

⁷³ Ibid., p. 15.

⁷⁴ "Biofuels: 1st, 2nd, and 3rd Generation," June 30 2015, post on blog "Open Clean Tech," <http://resources.opencleantech.com/our-blog/biofuels-1st-2nd-and-3rd-generation>, accessed October 2017.

⁷⁵ Ibid.